WORKED SOLUTION

STEP II Specimen paper, Question 8

Pure • Orders in Modular Groups

STEP II Specimen paper, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q8

The key idea

Modulo N, the number $N-1$ is just -1, and $m+n=N$ says $n \equiv -m$. So every power of n is a power of m up to a sign: $n^k \equiv (-1)^k m^k$. Whether k is even or odd decides whether that sign matters, and the three parts are the three ways the parities of the orders can combine. The final part then uses these facts to find orders without computing every power.

Solution

Throughout, $\equiv$ means congruent modulo N. We take the identity of G to be 1, as the question intends. (In fact this is forced whenever $N-1 \in G$: if e is the identity then $e(N-1) \equiv N-1$, that is $-e \equiv -1$, so $e=1$.)

The order of $N-1$

We have

$$(N-1)^2 = N^2 - 2N + 1 \equiv 1,$$

and $N-1 \neq 1$ because $N \geq 3$. So $(N-1)^1$ is not the identity but $(N-1)^2$ is: the order of $N-1$ is 2.

Setting up the parts

Since $m+n=N$, we have $n \equiv -m$, and so for every positive integer k

$$n^k \equiv (-1)^k m^k, \qquad \text{equivalently} \qquad m^k \equiv (-1)^k n^k.$$

Also $N-1 \equiv -1 \equiv n\,m^{-1}$ lies in G, because G is a group containing n and m^{-1}. So -1 is an element of G, and $-1 \not\equiv 1$ since $N \geq 3$. We shall also use the standard fact that if x has order r and $x^k = 1$, then $r \mid k$ (write $k = ar + b$ with $0 \leq b < r$; then $x^b = 1$, so $b = 0$ by minimality of r).

Part (i)

Let p and q both be even. Then

$$n^p \equiv (-1)^p m^p = m^p \equiv 1,$$

so $q \mid p$. In the same way $m^q \equiv (-1)^q n^q = n^q \equiv 1$, so $p \mid q$. Two positive integers dividing each other are equal, so $p = q$.

Part (ii)

Let p be even and q odd. As p is even, $n^p \equiv m^p \equiv 1$, so $q \mid p$. Also $2q$ is even, so

$$m^{2q} \equiv n^{2q} = (n^q)^2 \equiv 1,$$

giving $p \mid 2q$. Write $p = kq$; then $kq \mid 2q$, so $k \mid 2$ and $k = 1$ or 2. But $k = 1$ would make $p = q$ odd, which it isn't. Hence $k = 2$ and $p = 2q$.

? **Ask yourself** Why can't we simply argue $m^q \equiv -n^q \equiv -1$, so "$p$ must be $2q$"?

That calculation is correct and shows $m^q \not\equiv 1$ while $m^{2q} \equiv 1$, but on its own it only tells us $p \mid 2q$ and $p \nmid q$; it does not pin p down (for instance p could a priori be 2 with $q = 3$). The extra divisibility $q \mid p$, which comes from p being even, is what closes the gap.

Part (iii)

Suppose, for contradiction, that p and q are both odd, and let $r = pq$, which is odd. Then

$$n^r = (n^q)^p \equiv 1, \qquad \text{but also} \qquad n^r \equiv (-1)^r m^r = -(m^p)^q \equiv -1.$$

So $1 \equiv -1$, that is $N \mid 2$, which is impossible as $N \geq 3$. Hence p and q cannot both be odd.

(By symmetry, part (ii) also gives $q = 2p$ when q is even and p odd. So in every case the orders of m and $N - m$ are either equal and even, or one is twice the other, which is odd.)

The group modulo 21

Successive powers of 2 and 5, reduced modulo 21, are

k	1	2	3	4	5	6
2^k	2	4	8	16	11	1
5^k	5	4	20	16	17	1

(for example $2^5 = 32 \equiv 11$, $5^2 = 25 \equiv 4$, $5^4 \equiv 4^2 = 16$, $5^5 \equiv 80 \equiv 17$). The first 1 appears at $k = 6$ in both rows, so 2 and 5 both have order 6.

Now read off orders from the table: if x has order 6, then x^k has order $6/\gcd(k, 6)$. So

- $11 = 2^5$ and $17 = 5^5$ have order 6;

- $4 = 2^2$ and $16 = 2^4$ have order 3;
- $8 = 2^3$ and $20 = 5^3$ have order 2 (as $20 = N - 1$, the first part already tells us this);
- 1 has order 1.

That leaves 10, 13 and 19, each of which pairs with a known element to make 21:

- $13 = 21 - 8$, and 8 has even order 2. By (iii) and the remark after it, the order of 13 is either 2 (equal and even) or 1 (half of 2). It isn't 1, since $13 \neq 1$, so 13 has order 2. (Check: $13^2 = 169 = 8 \cdot 21 + 1$.)
- $19 = 21 - 2$ and $10 = 21 - 11$, and 2 and 11 have order 6. So each of 19 and 10 has order 6 or 3.

So every element of G has order at most 6. The cyclic group of order 12 has a generator, an element of order 12, and an isomorphism preserves orders of elements. Since G has no element of order 12, it is not isomorphic to the cyclic group of order 12.

There is a second way to finish: G has three elements of order 2, namely 8, 13 and 20, while a cyclic group of order 12 has exactly one (if g generates it, g^k has order 2 only for $k = 6$).

Checks

A computer check finds the full list of orders in G: 1 has order 1; $8, 13, 20$ have order 2; $4, 16$ have order 3; and $2, 5, 10, 11, 17, 19$ have order 6. (Directly, $10^3 = 1000 \equiv 13$ and $19^3 \equiv (-2)^3 = -8 \equiv 13$, so neither has order 3.) The counts $1 + 3 + 2 + 6 = 12$ match the size of G. Parts (i)–(iii) were also tested on every pair m, $N - m$ in every subgroup of the units generated by at most two elements, for all $N < 60$: nearly 1900 pairs, with no exceptions.

Common slips

- Assuming $N - 1 \in G$ without saying why. It is not given, but it follows from $m, n \in G$, since $N - 1 \equiv n\,m^{-1}$.
- In (ii), proving only $p \mid 2q$; one more divisibility is needed to get equality.
- Taking the identity for granted in weird cases: a subset such as $\{3\}$ modulo 6 is closed under multiplication with identity 3, and with $m = n = 3$ it would violate (iii). The argument relies on the identity being 1, which is clearly what is meant.

Beyond STEP

G is the group of units modulo 21, and by the Chinese remainder theorem it is isomorphic to the product of the unit groups modulo 3 and modulo 7, that is $C_2 \times C_6$: this is why it has three elements of order 2 and nothing of order 12. Deciding exactly

when the units modulo N form a cyclic group (precisely for $N = 2, 4, p^k, 2p^k$ with p an odd prime, the cases with a "primitive root") is a classic result of first-year number theory. Davenport's *The Higher Arithmetic* covers primitive roots, and Allenby's *Rings, Fields and Groups* treats the group theory.

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